

Udacity Programming Assignment 3 // Angela Gross // Parallel Programming

The screenshot displays the Udacity course viewer interface for the course "Intro to Parallel Programming". The browser address bar shows the URL: <https://www.udacity.com/course/viewer#!/c-cs344/l-86719951/e-90713805/m-90246143>. The left sidebar contains navigation links: DASHBOARD, CLASSROOM (active), MATERIALS, DISCUSSION, and OVERVIEW. The main area features a code editor with tabs for "student_func.cu", "reference_calc.cpp", "reference_main.cpp", and "reference_hw3.cu". The "student_func.cu" tab is active, displaying C++ code for a CUDA kernel. Below the code editor are buttons for "Reset", "Test Run", "Review Instructions", "Submit", and "Continue". A message box at the bottom states: "Good job! Your image matched perfectly to the reference image. Your program ran and executed in 2.310432 ms." On the right side, there is a "Correct!" notification, a "Get Help" section with links to discussion topics like "Ideal method to find max and min value from logLuminance array 3d ago", and a "Start a Discussion" button.

Udacity Course Viewer Interface

Breadcrumbs: Intro to Parallel Programming > Classroom - Udacity > Dijkstra's algorithm > MightyText > Course: ST: Apps of Parallel... > Home

Address Bar: <https://www.udacity.com/course/viewer#!/c-cs344/l-86719951/e-90713805/m-90246143>

Sidebar Navigation:

- DASHBOARD
- CLASSROOM (Active)
- MATERIALS
- DISCUSSION
- OVERVIEW

Code Editor Tabs: student_func.cu | reference_calc.cpp | reference_main.cpp | reference_hw3.cu

```

1 #include "utils.h"
2
3 // =====
4 // student_func.cu
5 // -----
6 // Angela Gross
7 // CSCI-595: Parallel Programming
8 // Homework 3
9 // Spring 2015
10 // -----
11 // This transforms the log of the luminance channel of an image by compressing its range to [0, 1], and this
12 // by making a cumulative distribution of these luminance values. This is done in parallel by implementing a
13 // histogram, and finally an inclusive prefix sum on those values (all using CUDA).
14 //
15 // In addition to the forums on the Udacity course, the following websites assisted in my implementation:
16 // - http://www.cuvilib.com/Reduction.pdf
17 // - http://developer.nvidia.com/GPUGems3/gpugems3_ch39.html
18 // =====
19
20 ////////////////////////////////////////
21
22 // TYPEDEF AND OPERATIONS FOR REDUCTIONS
23 typedef float (*op_func) (const float &, const float&);
24 
```

Action Buttons: Reset | Test Run | Review Instructions | Submit | Continue

Feedback Section:

- Correct!** Please see detailed feedback in the output area below.
- Get Help**
 - Ideal method to find max and min value from logLuminance array 3d ago
 - HW3: Intermittent success Mar 9
 - HW3: Can Hillis-Steele be used for exclusive scan? Mar 7
 - Unspecified Launch Error in CUDA Mar 7
 - HW3: confused by parameters to your_histogram_and_prefixsum() Mar 3
- [Start a Discussion](#)
- [Report an Issue](#)

Status Message: Good job! Your image matched perfectly to the reference image
Your program ran and executed in 2.310432 ms.